



Specialty Gas Distribution Systems Best
Known Methods (BKMs) Guidelines

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Item	Description
1 General Information	
1.1 Document Name	Specialty Gas Distribution Systems BKMs guidelines
1.2 Document Number	CSG-2
1.3 Description	Equipment that consists of fluid handling pipelines and components made of stainless steel, specifically designed for high-purity applications and used to supply gas to utilization networks with sustainable pressure and flow.
1.4 Owner	stamenghi
1.5 Applicable FPM System IDs	CSG/BSG
1.6 Revision History	0
1.7 Glossary of terms	http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef83a1f41c
2 Minimum Recommendations	Hide or Display Rows as Needed
2.1 Mechanical	
2.1.1 Design Parameters	1. This document applies to all specialty gas and liquid chemical distribution routing from the gas supply system to the final point of connection (LPOC). Depending on application, the POC can be located as follows: 1.1 For Hazardous production materials (HPM) gases: in the Valve manifold box (VMB) stick, except for those applications where a point to point distribution is required, such as WF6, SiCl4 etc. 1.2 For non HPM gases: on the lateral and/or valve manifold panel (VMP) stick. 2. Piping shall be routed through designated trenches on pipe racking made of metal structures or "UNISTRUT" bars, with all piping hung by mechanical cushioned supports. Piping must be exposed to allow for visual inspection of its entire length. The routing shall be designed based on space management and the manufacturing area layout. 3. The system design shall conform to all applicable (site-specific) local codes and regulations. The list below includes, but is not limited to, some of the typical codes and regulations: 3.1 International building code (IBC) 3.2 International fire code (IFC) and local/country fire regulations 3.3 Occupational Safety and Health Administration (OSHA) 3.4 Environmental Protection Agency (EPA), and other local/country toxic gas regulations 3.5 International Mechanical Code (IMC), country mechanical/installation regulations 3.6 SEMI S2, JIS, CGA and other country gas regulations 3.7 Other local/country regulation for compressed gases, storage and distribution codes, and welding certification 3.8 Corporate safety requirements, Factory Mutual (FM) requirements, and risk assessment in "Micron Safety Standard for Gas monitoring" (Active link in section 6.2.2). 4. HPM distributions shall be equipped with an appropriate Toxic Gas Monitoring (TGM) system, according to the "Safety Standard for Toxic Gas monitoring" (Active link in section 6.2.2). 5. Gas distribution systems shall be designed and installed with compatible materials and components to guarantee unaltered purity and properties to the POC. The welding techniques utilized and the installation methods used in the fabrication of the systems, shall be declared by manufacturers and installer(s) and shall conform to the "UHP Stainless Steel BKMs guidelines" (Active link in section 6.2.4). 6. Pipelines shall be welded using an orbital welding technique, with Ar as an inert environment for tubing and the weld head. VMBs and other equipment components shall be coupled with high vacuum fittings made of face seal elements mechanically coupled with a metal gasket (VCR) connection, except for dedicated application specified in the design phase.
2.1.2 Operating Parameters	1. Compressed and compressed liquefied gases are distributed at ≤ 75psig. Liquefied low vapor pressure gases are distributed in a range between -10psig and 25psig, depending on the specific vapor pressure at room temperature. Liquid chemicals are supplied at delivery pressure < 75psig by inert gas pressurization. 2. Specialty gases shall be distributed at room temperature, between 68°F and 77°F (20°C and 25°C), based on the site-specific operational parameters. 3. Low vapor pressure gases may need to be operated with a dedicated temperature control system, to increase capacity and prevent gas condensation in the distribution system.

2.1.3 Design Configuration	1. Gas distribution systems shall be designed according to gas properties and the safety classification as outlined in the "Specialty gas systems concept design" (Active in the link 6.2.1) and "Micron Safety Standard for Gas monitoring" (Active link in section 6.2.2). 2. HPM gases shall be distributed through "nodes". Depending on gas properties, quantity of supplied tools, gas usage, the distribution system shall be designed, but not limited, with: 2.1 Main pipeline building isolation cabinet(s) (BIC), multiple VMBs in the outdoor bulk systems 2.2 Main pipeline distribution valve box (DVB), multiple VMBs, in the N+1 indoor systems 2.3 Main pipeline VMBs, in the individual indoor systems 3. Non HPM gases shall be distributed by main pipeline(s), lateral(S), and sub-lateral(s). 4. HPM gas distributions shall be designed with following configuration and nodes: 4.1 BICs shall be designed with sticks having double isolation process IN, purge IN, excess flow switch, pressure regulator and transducer, line filter, vent OUT, stick isolation valve. The BICs shall be controlled by a HMI and PLC/microprocessor for automatic sequences and LSS interlock. 4.2 DVBs shall be designed with dual, double isolation valve (air operated valves (AOV) + manual) process IN, pressure transducer, double isolated sticks. The manifolds shall be all welded, except for the process OUT which shall have a VCR connection for the VMB supply. AOVs shall be interlocked by the LSS, via a dedicated controller and/or the gas supply system controller. 4.3 VMBs shall be designed with dual, double isolation valve process IN, pressure transducer, double isolated sticks. AOVs shall be interlocked by the LSS, via a dedicated controller and/or the gas supply system controller. 5. The HPM nodes shall be equipped with air operated valves (AOV) and manual valves to provide full isolation and Life Safety System (LSS) interlock. Valves shall be rated for pressure range ≤ 250psig, and flow capacity ≥ 0.25 Cv as minimum. 6. The HPM delivery pressure in the nodes shall be monitored by pressure transducers with 100psig range, except for dedicated applications as directed during the design phase. 7. Liquid chemicals shall be distributed in a liquid state from the clean sub-fab and pushed to the tool with an inert gas, He, or N2. 8. Non HPM distribution shall be designed with main/lateral pipelines and equipped with expansion valves and POC valves located in the proximity of the tools location. Valves shall be rated for pressure range ≤ 250psig, and flow capacity ≥ 0.25 Cv as a minimum.
2.1.4 Capacity	1. Main distribution nodes (HPM) and laterals (non HPM) shall be designed to ensure flow capacity with maximum pressure loss of ≤ 2psi at the end of line (EOL). 2. All material and components shall be rated for 250psig maximum pressure, except for dedicated applications directed during the design phase. 3. All components shall be rated with appropriate flow capacity, as indicated by manufacturers with the component "Cv" and able to meet/exceed the needed capacity identified in the section 2.5.4. 4. The pressure regulator capacity shall be identified based on specific "flow curve".

Item	Description
2.1.5 Redundancy	1. HPM gases main distribution to DVB in the N+1 configuration as indicated in the "Specialty gas concept Design" (Active link in section 6.2.1)
2.1.6 Expansion	and dedicated applications as directed during the design phase 1. HPM gases shall be equipped with expandable nodes. 2. Non-HPM systems shall be equipped with expansion valves at the end of the main distribution pipeline, as a minimum. Additional expansion valves shall be identified along the pipe route for expected future expansion of the distribution.
2.1.7 Operation	1. Distribution systems shall be operated according to the site and/or gas management system O&M procedures.
2.1.8 Maintenance	1. Gas distribution systems shall be maintained according to the site and/or gas management system O&M procedures.
2.1.9 Materials	1. Materials utilized in the specialty gas distribution systems shall be suitable for SS high purity installation according to the requirements identified in the "UHP Stainless Steel BKM's guidelines" (Active link in section 6.2.4).
2.1.10 Components	1. Materials utilized in the specialty gas distribution systems shall be suitable for SS high purity installation according to the requirements identified in the "UHP Stainless Steel BKM's guidelines"(Active link in section 6.2.4).
2.1.11 Manufacturers	1. Air Products, Air Liquide, Linde, Norcimbus, KCtech, TSNC, TYK
2.1.12 Installation	1. Specialty gas distribution systems shall be installed by personnel specifically trained and certified for high purity systems, including the welding and assembly, and testing and qualification processes. 2. For installations performed by external companies, those companies shall provide all documentation related to the training and certification of all personnel involved in welding, assembly, testing, and qualification processes of stainless steel high purity systems using the execution requirements contained in the "UHP Stainless Steel BKM's guidelines" document (Active link in section 6.2.4).
2.1.13 QA/Testing	1. Gas distribution systems shall be finally qualified based on, but not limited to, the following tests: 1.1 Leak test and/or pressure test to certify the mechanical integrity of welding and VCR connections. 1.2 Moisture content to certify the complete dry out of the interior of the pipeline to avoid any abnormal reaction with the specialty gas. 1.3 Particle test to certify the cleanliness and good quality of the welds and ensure that any particle contamination does not exceed the maximum accepted level. 2. Each site shall develop its own construction procedure and Quality Assurance/Quality Control (QA/QC) procedures with detailed descriptions of the documents for historical archiving.
2.1.14 Other	1. Gas distribution systems shall be documented based on, but not limited to, the following suppliers documents: 1.1 Layout distribution through fab building(s) and level(s), indicating wall penetration, pipeline sizing 1.2 Nodes location in the clean sub-fab (HPM) 1.3 Main/lateral pipelines with indication of isolation valves and POC valves 1.3 All documents supplied in printed and electronic versions 2. Test and commissioning report, final system acceptance, and hand over. 3. Toxic gas monitoring sampling point and detector wiring code for HPM gases
2.2 Electrical	
2.2.1 Design Parameters	1. All metallic components of a specialty gas distribution system shall be bonded together to form a common low impedance path to the electrical grounding system. 2. Appropriate wiring methods shall be used for the installation of a specialty gas distribution system.
2.2.3 Design Configuration	1. HPM gas distributions, VMBs, and related point of connection boxes shall be equipped with gas detection. TGM shall be in compliance with the "Safety Standard for Toxic Gas monitoring" (Active link in section 6.2.2) and local codes and regulations. 2. For automated VMBs utilized for HPM gases, the PLC shall be constructed per UL508a or an equivalent listing agency.
2.3 I & C	
2.3.1 Design Parameters	1. HPM gas nodes shall be equipped with AOV to shut down the gas supply according to the LSS interlock strategy. 2. Programmable Logic Controller (PLC) and Human Machine Interface (HMI) shall be provided with sequences of operation, alarm thresholds and loop tuning parameters. 3. Double-contained distribution systems shall be equipped with, but not be limited to, the following elements: 3.1 An indicator pressure switch (IPS) shall monitor the double containment and initiate a SCADA alarm if there is a loss of vacuum or pressure. 3.2 A pressure transmitter (PT) shall monitor/control the outer containment and initiate a SCADA alarm if there is a loss of vacuum or pressure.
2.3.3 Design Configuration	1. In HPM gas distribution systems with automated nodes, PLC cabinets shall be constructed per UL508a or an equivalent listing agency. 2. Delivery pressure shall be monitored with the SCADA system and related data returned for trending and troubleshooting.
2.3.7 Operation	1. The HMI of HPM gas nodes shall be capable of providing access to all operational and alarm parameters.
2.3.8 Maintenance	1. For HPM gases preventative maintenance shall include, but is not limited to, visual PLC and instrumentation inspection, power supply inspection, and PLC lithium battery inspection and/or replacement.
2.3.10 Components	1. All instruments and control components shall be listed by UL, CE, or equivalent agencies.
2.3.13 QA/Testing	1. Functional Acceptance Tests (FATs) shall be performed and documented. FATs shall include, but are not limited to, PLC interlock performance, HMI functionality, alarm functionality, and instruments and sequence of operation compliance.
2.3.14 Other	1. PLC and HMI software programs, I/O lists and any other applications including SCADA shall be developed and then delivered to the Micron Controls team. When external vendors develop the software, updated versions of the software, with all comments, shall be delivered to the Micron Controls team.
2.4 Chemical	

2.5 Gas	
2.5.1 Design Parameters	1. The systems concept design shall be identified based on, but not limited to, the criteria outlined in the "Specialty gas concept Design" document (Active link in section 6.2.1). 2. The project team involved in designing a new specialty gas distribution system shall include team members from the Operation & Maintenance and Safety organizational models identified at each site. 3. Main design parameters include, but are not limited to, the following items: 3.1 POC delivery pressure: the required pressure at the tool point of connection (POC), ≤ 75psig, except for specific requirements. 3.2 Peak flow demand: the flow demand when all tools connected to the system require gas simultaneously. 3.3 Average flow demand: the average flow of the system on a daily, weekly, and monthly basis, according to existing gas usage in the same site or other sites, comparable for technology and manufacturing device. 3.4 Design flow: the flow rate determined as the sustainable flow rate according to the average flow rate data provided by TSRs and/or any other details by the supplier's modeling program. 3.5 Pipe size: the size of main and lateral pipelines as well as the nodes supply in the HPM gas applications. 3.6 Pressure loss: the expected pressure loss at the EOL and the farthest LPOC. It is differentiated based on two main types of distribution systems: 3.6.1 Outdoor systems (BSGS): ≤ 5psi 3.6.2 Indoor systems (gas cabinet/panels etc.): ≤ 2psi 3.7 Toxic Gas Monitoring requirements: based on HPM gases safety requirements included in the "Safety Standard for Toxic Gas monitoring" (Active link in section 6.2.2). 3.8 Tool quantity: total number of tools to be supplied by the system. 3.9 Gas area sharing/loading: the sharing of a system to different production areas to restrict the potential impacts in the event of a system shut down and/or contamination. 4. Non–HPM gas distribution systems shall be designed with a quantity of LPOC valves (one as a minimum) to supply the required tools and 20% of additional capacity. HPM gas distribution systems shall be designed with required VMB stick capacity and future expansion capability.
2.5.2 Operating Parameters	1. The delivery pressure shall be identified based on the tool requirements and the recommended guidelines indicated in the "Operating systems guidelines" document (Active link in section 6.2.3).
2.5.3 Design Configuration	1. Gas distribution systems shall be designed to assure the required delivery pressure and flow rate with unaltered quality. The pipeline sizing shall be assisted by capacity modeling using "Pipe–flo", or an alternative equivalent modeling program. 2. Nodes for HPM gases shall be designed to assure the required flow rate and additional capacity for future changes. Valves with Cv 0.25 shall be used as a minimum. Higher valve Cv will be used when supplying low vapor pressure gas or when higher demand is expected in the future. 3. Pressure regulators in the BIC(s) shall be designed with free poppet configuration, and sized according to individual flow curve. 4. Excess flow switches in the BIC sticks shall be sized to a trip point corresponding to a large leak, similar to the pipeline breaking Trip points will be set with a delay time comprised between 5 and 10 seconds. 5. Filters in the BIC sticks shall be rated for 0.003 μ particle retention, sized to guarantee ≤ 2psi pressure loss at the maximum designed flow rate. 6. The HPM gas nodes shall be designed with dual process IN, to allow for N+1 configuration, as well as remodeling capability.

Item	Description
2.5.4 Capacity	1. The system capacity shall be identified based on, but not limited to, the following: 1.1 Pipelines. Design flow rate with 30% over capacity as minimum 1.2 BIC and DVB. Design flow rate with 30% over–capacity as minimum and 20% additional sticks, one as minimum, based on project requirement 1.3 VMB. Design flow rate with 30% over–capacity as minimum 1.1 Non HPM distribution. Based on design flow rate and 30% over–capacity. The quantity of valves shall be determined according to the tools, with 20 % over–capacity, one as minimum. 2. Valves Cv. 0.25 as minimum in all applications. Cv 0.5 for higher flow rate applications, as directed in the design phase. 3. Pressure regulators. Based on design flow and regulator flow curves. Identified regulators shall be able to meet the design flow rate with 30 % over capacity, with 70psig incoming pressure and 40psig out coming pressure.
2.5.5 Redundancy	1. N+1 configuration for HPM gases
2.5.6 Expansion	1. BICs configuration in the BSGS systems, DVB configuration in the N+1 HPM gases, main/lateral expansion valve in non HPM gas distribution
2.5.7 Operation	1. The gas distribution systems shall be operated according to the equipment manufacturer O&M manual, site and/or the Total Gas Management group's operating procedures. 2. Gas distribution equipment such as BICs, DVBs, and VMBs shall be monitored routinely, either in the sub–fab or the Control Room from SCADA. Delivery pressure shall be monitored once a day as a minimum. 3. Any operation for maintenance and/or troubleshooting shall be recorded and retained for excursions, and manufacturing support. 4. A software program shall be used to record abnormal events for troubleshooting and lessons learned. 5. Gas operation groups, as part of the Facilities team or the Total Gas Management group, shall release and maintain the details for: 5.1 Turn on/off of specialty gases in each node of the distribution 5.2 Defacilitation of equipment and pipelines. 6. Operations team shall keel and maintain the complete list of gas distribution layouts.
2.5.8 Maintenance	1. The gas distribution equipment shall be maintained according to the manufacturer's O&M manual, site and/or the Total Gas Management group's operating procedures. 2. A procedure shall be implemented to quickly respond to a gas contamination, which can occur due to impurities contained in the gas source; incorrect operations at the tools; or material and component failure. This procedure should include, but not be limited to, the following items: 2.1 A notification process from Manufacturing team members to Facilities Operations, escalating as the severity of the issue increases to those with increased responsibility. 2.2 SCADA data log system and any other log–book checks for typical operating delivery pressure at each distribution node. 2.3 Coordinate the actions with tool owners of tests to re–qualify new gas and restart the supply to all production area(s). 2.5 Record and share the event by system in place at each site. 3. Repair and maintenance of specialty gas systems shall be performed by specially–trained personnel to operate UHP SS systems. Training program shall be recorded and updated routinely according to the site program. 4. Repair and/or replacement of components shall be carried out according to the gas system manufacturer, Operations, and Total gas Management group procedures. Whenever possible, the failing components shall be sent back to the manufacturer for a failure analysis and results shared within the company.
2.5.9 Materials	1. The materials utilized for maintenance shall be the same of those installed in a new system according to the manufacturer's material list or equivalent, based on Micron approval.
2.5.10 Components	1. The components utilized for maintenance shall be the same of those installed in a new system according to the manufacturer's material list or equivalent, based on Micron approval.
2.5.11 Manufacturers	1. Air Products, Air Liquide, Linde, Norcimbus, KCTech, TSNC, TYK, and their subsidiaries
2.5.12 Installation	1. Air Products, Air Liquide, Linde, Norcimbus, KCTech, TSNC, TYK, their subsidiaries; and all approved site contractors for UHP Stainless Steel installations.
2.5.13 QA/Testing	1. Gas distribution systems shall be 100% tested and commissioned according to the manufacturer's specifications and site procedures. 2. The qualification shall be performed to guarantee the system integrity and shall include, but not be limited to, the following: 2.1 Leak and/or pressure test executed according to Micron procedures and local codes and regulations. 2.2 Moisture analysis to confirm that the pipeline/system has been properly installed, purged, and conditioned. 2.3 Particle test to confirm that the entire system has been welded, purged, cleaned, and conditioned properly. 3. All other quality requirements and checks related to material selection and inspection, weld techniques and coupon traceability, and installation sequence shall be followed based on each site's specification and "UHP Stainless Steel BKM's guidelines" (Active link in section 6.2.4).
2.5.14 Other	
2.6 Water & Waste Water	
2.7 Other	
3 Best Practices	Hide or Display Rows as Needed

3.1 Mechanical			
3.2 Electrical			
3.3 I & C			
3.4 Chemical			
3.5 Gas			
3.6 Waste & Waste Water			
3.7 Other			
4 Supporting Documentation			
4.1 Sequence of Operations			
4.2 System Acceptance Criteria			
4.3 Design Concept			
4.4 P & IDs			
4.5 Safety			
4.6 Block Diagrams			
4.7 Other Technical Document			
5 Approved Variances	[Place holder – Add link to variances]		
5.1			
5.2			
5.3			
6 Other Links			
6.1 Definitions			
6.2 References	1. Specialty gas concept design – http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef8656df28		
	2. Safety standard for toxic gas monitoring – http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef8303e592		
	3. Specialty gas systems Operation guideline – http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef82ffb084		
	4. UHP Stainless Steel BKMs guidelines - http://edm.micron.com/cgi-bin/mtgetdoc.exe?itemID=09005aef867020b3		
6.3 Industry Standards			
6.4 Procedures			
6.5 Interfacing System Standards			
7 Notes			
8 Revision History			
0	9/25/2015 Initial issue		
9 Appendix			
	Term/Acronym	Definition / Description	
	FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs	
	GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards	
	GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities	
	Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team	
	Global Facilities Construction Team	This is used interchangeably with Global Construction Team	
	Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team	
	Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team	